Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

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Question 4(6)

FP Tree:-

TID	List of Items
T100	I_1, I_2, I_5
T200	I_2, I_4
T300	I_2, I_3
T400	I_1, I_2, I_4
T500	I_1, I_3
T600	I_2, I_3
T700	I_1, I_3
T800	I_1, I_2, I_3, I_5
T900	I_1, I_2, I_3

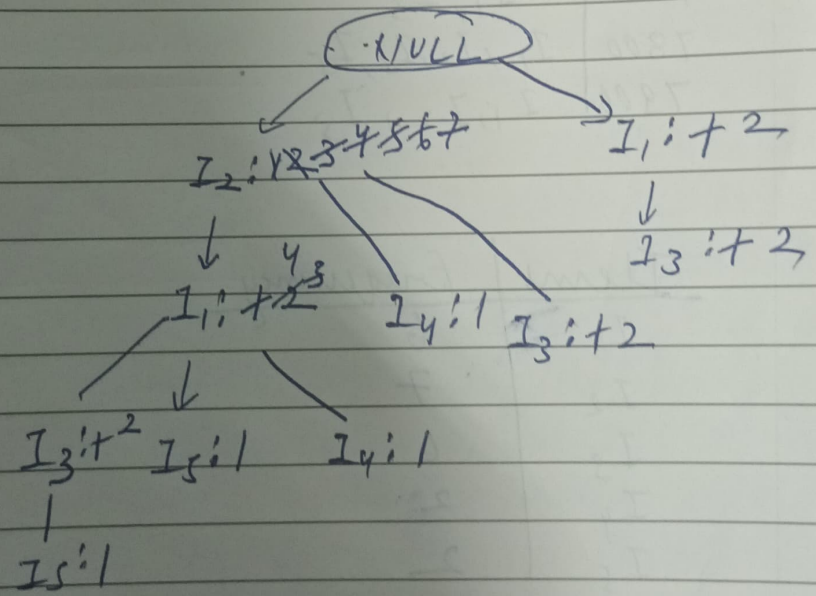
Items	Frequency
I_1	6
I_2	7
I_3	6
I_4	2
I_5	2

Min Support = 2

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

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TID	Items	Ordered IS
T100	$I_1\ I_2\ I_5$	$I_2\ I_1\ I_5$
T200	$I_2\ I_4$	$I_2\ I_4$
T300	$I_2\ I_3$	$I_2\ I_3$
T400	$I_1\ I_2\ I_4$	$I_2\ I_1\ I_4$
T500	$I_1\ I_3$	$I_1\ I_3$
T600	$I_2\ I_3$	$I_2\ I_3$
T700	$I_1\ I_3$	$I_1\ I_3$
T800	$I_1\ I_2\ I_3\ I_5$	$I_2\ I_1\ I_3\ I_5$
T900	$I_1\ I_2\ I_3$	$I_2\ I_1\ I_3$



Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks | Question No.

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Items	Conditional Pattern Base	Conditional Frequent Item tree	Frequent Pattern generated
I_5	$\{I_2, I_1, I_3 : 1\}$ $\{I_2, I_1 : 1\}$	$I_2, I_1 : 2$	$I_2 I_5 : 2, I_1 I_5 : 2$ $I_2, I_1 I_5 : 2$
I_4	$\{I_2 I_1 : 1\}$ $\{I_2 : 1\}$	$I_1 : 2$	$I_2 I_4 : 2$
I_3	$\{I_2, I_1 : 2\}$ $\{I_2 : 2\}$ $\{I_1 : 2\}$	$I_2 : 4, I_1 : 2$	$I_2 I_3 : 4, I_1 I_3 : 2$ $I_2 I_1 I_3 : 2$
I_1	$\{I_2 : 4\}$	$\{I_2 : 4\}$	$I_2 I_1 : 4$

Question 4 (a)

→ Conside the data of 4(6) Support 2 confidence 50%

1 Item Set	Frequency
I_1	6
I_2	7
I_3	6
I_4	2
I_5	2

Frequent 1-Itemsets	Frequency
I_1	6
I_2	7
I_3	6
I_4	2
I_5	2

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.

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Frequent 2 Item Sets	Frequency
I_1, I_2	4
I_1, I_3	4
I_1, I_5	2
I_2, I_3	4
I_2, I_4	2
I_2, I_5	2

Frequent 3-item Sets	Frequency
I_1, I_2, I_3	2
I_1, I_2, I_5	2

Frequent 4 is not possible.

→ Frequent 3-Items Sets =
$I \Rightarrow \{1, 2, 3\}$ and $\{1, 2, 5\}$

Min-Support = $2/9 = 22.22\%$

Min-Confidence = 50%

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

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⇒ For I ⇒ {1, 2, 3}

Rule 1 :- {1} → {2, 3}

Confidence = Support {1, 2, 3} / Support (1)

= 2/9 = 33.34%

6/9 < 50%

Invalid.

Rule-2 :- {2} → {1, 3}

Confidence = 28.57% < 50%

Invalid

Rule-3 :- {3} → {1, 2}

Invalid.

Rule-4 :- {1, 2} → {3}

Valid

Rule 5 :- {1, 3} → {2}

Valid.

Rule 6 :- {1, 2} → {1}

Valid.

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.	START WRITING HERE
	⇒	For 1 ⇒ {1, 2, 5}
		Rule 1:- {1} → {2, 5} Invalid
		Rule 2:- {2} → {1, 5} Invalid
		Rule 3:- {5} → {1, 2} Valid.
		Rule 4:- {1, 2} → {5} Valid.
		Rule 5:- {1, 5} → {2} Valid.
		Rule - 6:- {2, 5} → {1} Valid.

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.

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Question-2(b)

Data Points:- {2, 4, 10, 12, 3, 20, 30, 11, 25}

Distance func^{c}:- Euclidean Distance

Initial centroid are $M_1 = 4$ $M_2 = 11$

Step 1:-

Data Points	Distance to M_1	Distance to M_2	Cluster	New cluster
2	2	9	1	
4	0	7	1	
10	6	1	2	
12	8	1	2	
3	1	8	1	
20	16	9	2	
30	26	19	2	
11	7	0	2	
25	21	14	2	

New Centroids:- $M_1 = 3$ $M_2 = 18$

Step-2:-

New Centroids $M_1 = 4.75$ $M_2 = 19.6$

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.

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Data Points	Distance to M_1	Distance to M_2	Cluster	New Cluster
2	1	16	1	1
4	1	14	1	1
10	7	8	2	1
12	9	6	2	2
3	0	15	1	1
20	17	82	2	2
30	27	12	2	2
11	8	7	2	2
25	22	7	2	2

Step-3:—

New Centroids:—

$M_1 = 7$

$M_2 = 25$

Data Points	Distance to M_1	Distance to M_2	Cluster	New Cluster
2	2.75	17.6	1	1
4	0.75	15.6	1	1
10	5.25	9.6	1	1
12	7.25	7.6	2	1
3	1.75	16.6	1	1
20	15.25	0.4	2	2
30	25.25	10.4	2	2
11	6.25	8.6	2	1
25	18.25	5.4	2	2

Step 4 :—

Data Points	Distance to M_1	Distance to M_2	Cluster	New Cluster
2	5	23	1	1
4	3	21	1	1
10	3	15	1	1
12	5	13	1	1
3	4	22	1	1
20	13	5	2	2
30	23	5	2	2
11	4	14	1	1
25	18	0	2	2

Final cluster are :—

$C_1 = \{2, 4, 10, 11, 12, 3\}$
$C_2 = \{20, 30, 25\}$

Question 2 (a)

Data visualization :— Data visualization is presenting the data in a graphical or pictorial format. Some of data visualization techniques are as follows :—

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

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	①	Pixel-oriented visualization technique :-
		In this technique, there is a separate subwindows for the value of each attributs and represented by one colored pixel.
	②	Geometric projection visualization techniques :-
		→ Scatterplot matrices. → Hyper slice → Parallel Coordinates.
		Scatterplot matrices composed of scatter plots of all possible pairs of variables in a dataset.
	③	Icon-based visualization techniques :-
		In this, each multidimensional data item is mapped to an icon. The most commonly used icon based techniques are :- a) Chernoff faces b) Stick figures.

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		Moderator	
		Re-Assessor	

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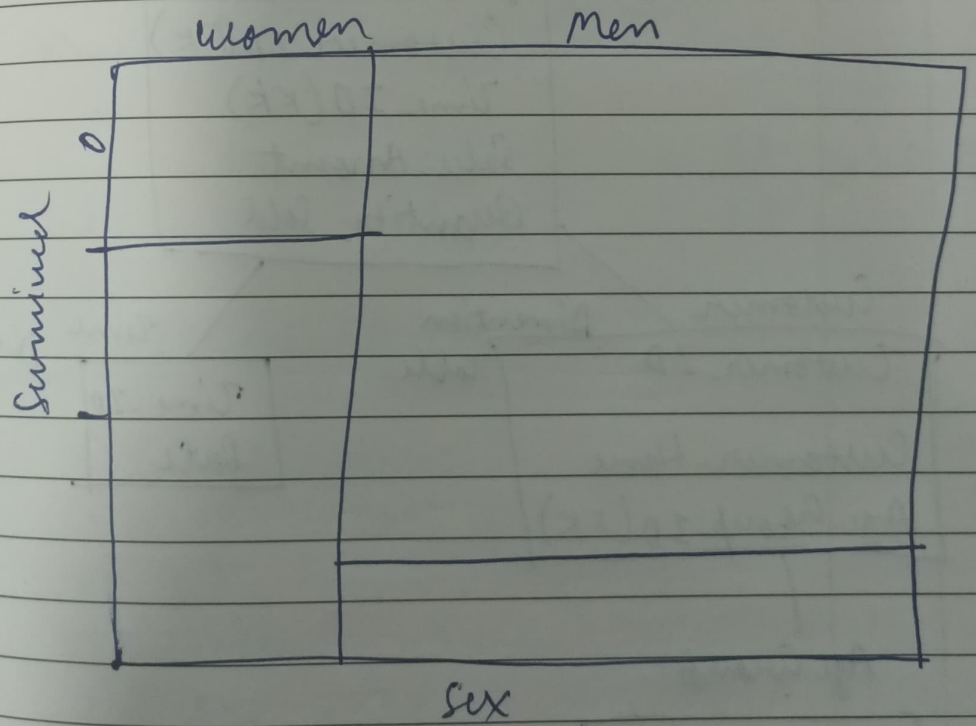
Hierarchical visualization techniques:-

→ Hierarchical technique partitions all dimensions in to subsit.

→ Some of the visualization techniques are:-

a) Dimensional Stacking
b) Mosaic Plot
c) Worlds - within worlds.
d) Tree map
e) Visualization complex data and relations.

e.g.



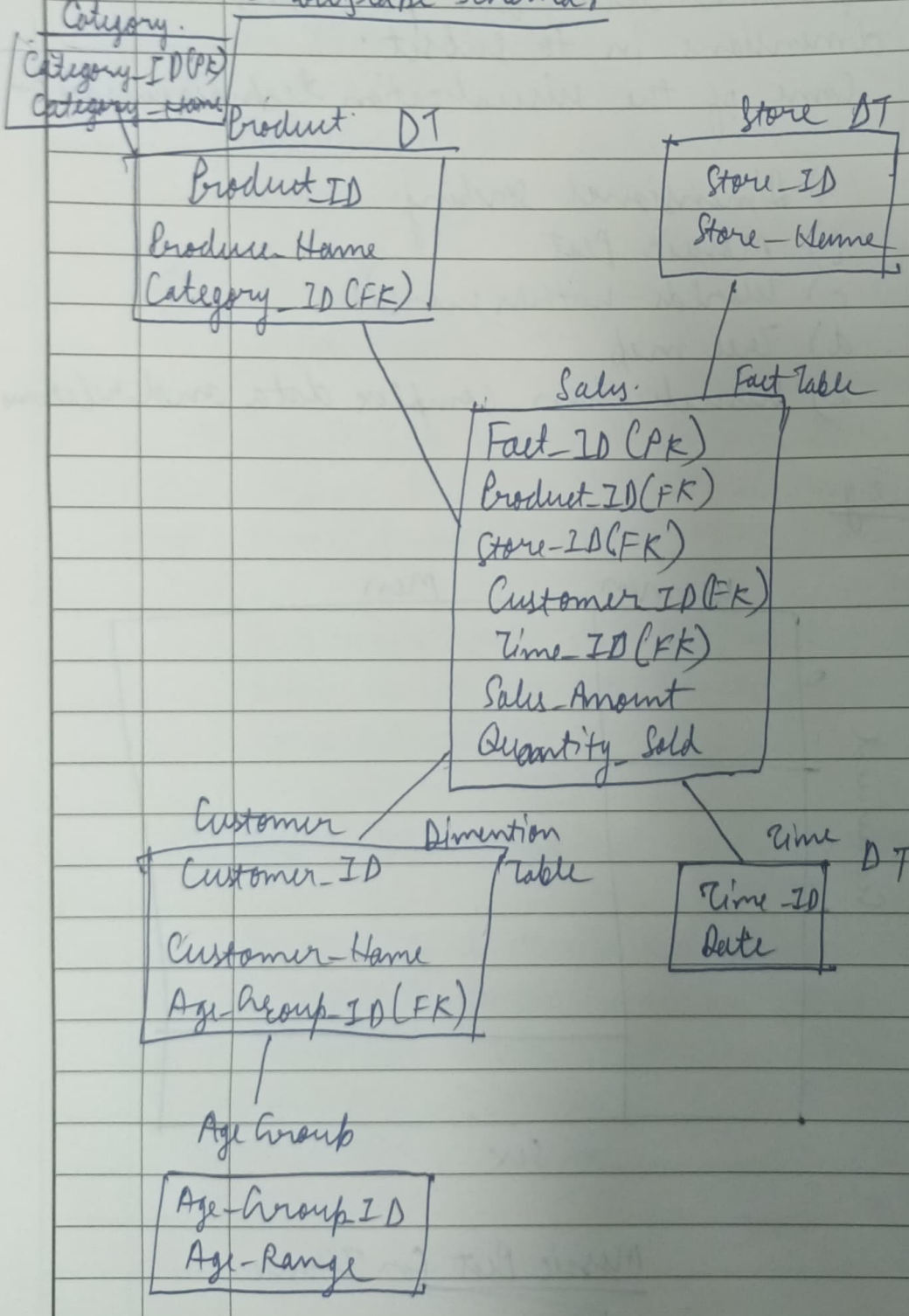
Mosaic Plot for Titanic

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		Moderator	
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Question 1 (a)

Snowflake Schema:-



Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.	START WRITING HERE

Question 1 (b)

Data of age :-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25,
30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

① **Min Max Normalization :-**

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$x = 35$
$x_{min} = 13$
$x_{max} = 70$

$$x_{norm} = \frac{35 - 13}{70 - 13} = \frac{22}{57} \approx 0.386$$

② **Z-Score Normalization :-**

$$Z = \frac{x - \mu}{\sigma}$$

$x = 35$
$\sigma = 12.94$
$\mu = \frac{810}{27} \approx 30$

$$Z = \frac{35 - 30}{12.94} = \frac{5}{12.94} \approx 0.386$$

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

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	(3)	**Normalization by Decimal Scaling:-**

$$X_{norm} = \frac{X}{10^k}$$

for $k=2$ value is less than 1.

$$X_{norm} = \frac{35}{10^2}$$

$$= 0.35$$

(4) Z score normalization is preferred as dataset has a relatively wide range of values from 13 to 70 and an outlier (70).

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.	START WRITING HERE

Question 1(c)

Benefits of a Data Lakes:

① **Handles diverse data:** – Can store structured, semi structured and unstructured data without preprocessing.

② **Scalability:** Easily scales to store large volumes of data cost-effectively.

③ **Flexibility:** Stores raw data, enabling future analytics and experimentations

④ **Cost Effective:** Lower storage costs compared to structured systems.

⑤ **Big Data Integration:** Works well with big data tools for advanced analytics.

Challenges of Data Lakes:

① **Data Governance Issues:** Lack of structure can lead to poor data quality and inconsistent formats.

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

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	(2)	Complex Management: Managing large volumes of unmonitored data can be difficult without proper metadata.
	(3)	Slow Querying: Performance can suffer due to raw, unprocessed data.
	(4)	Security Concerns: - Ensuring proper security and compliance can be challenging.

Total Marks of Question no.

Examiner
Moderator
Re-Assessor

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Question 3 (6)

	Yes	No
Yes	800	200
No	300	700

True Positive (TP) = 800
False Positive (FP) = 300
False Negative (FN) = 200
True Negative (TN) = 700

① **Accuracy :-**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{800 + 700}{800 + 700 + 300 + 200}$$

$$= \frac{1500}{2000}$$

$$= 0.75 \qquad 75\%$$

② **Error Rate :-**

$$\text{Error} = \frac{FP + FN}{TP + TN + FP + FN}$$

$$= \frac{300 + 200}{800 + 700 + 300 + 200} = \frac{500}{2000} = 0.25$$

25%

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.	START WRITING HERE

3- True Positive Rate / Recall / Sensitivity:-

$$TPR = \frac{TP}{TP + FN}$$

$$= \frac{800}{800 + 200}$$

$$= 0.8 \qquad 80\%$$

4- False Negative Rate (FNR):-

$$FNR = \frac{FN}{TP + FN}$$

$$= \frac{200}{800 + 200} = 0.2$$

$$20\%$$

5- False Positive Rate (FPR):-

$$FPR = \frac{FP}{FP + TN}$$

$$= \frac{300}{300 + 700}$$

$$= \frac{300}{1000} = 0.3 \quad 30\%$$

Total Marks of Question no.

Examiner

Moderator

Re-Assessor

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6- True Negativity Rate (TNR):—

$$TNR = \frac{TN}{FP + TN}$$

$$= \frac{700}{300 + 700} = 0.7$$

70%

7- Precision:—

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{800}{800 + 300}$$

$$= 0.727$$

72.7%

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.

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Question 3(a)

Income Level	Credit History	Loan Amount	Default
High	Good	Low	No
Low	Poor	High	Yes
Medium	Good	Medium	No
Low	Good	Low	No
High	Poor	Medium	Yes
Medium	Poor	High	Yes

Decision Tree (ID3) :-

a) Attribute :- Income Level

values (Income Level) = High, Low, Medium

Entropy of whole dataset :-

$S = [3+, 3-]$

$$\text{Entropy}(S) = -\frac{3}{6}\log_2\frac{3}{6} - \frac{3}{6}\log_2\frac{3}{6}$$

$$= 0.5 + 0.5$$

$$= 1$$

Total Marks of Question no.		Examiner	
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		Re-Assessor	

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$S_{High} = [1+, 1-]$

$Entropy(S_{High}) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}$

$= 1$

$S_{medium} = [1+, 1-]$

$Entropy(S_{medium}) = 1$

$S_{Low} = [1+, 1-]$

$Entropy(S_{Low}) = 1$

$Gain = Entropy(S) - \sum_{v \in (High, medium, low)} \frac{|S_v|}{|S|} Entropy(S_v)$

$Gain(S, IL) = 1 - \frac{2}{6} \times 1 - \frac{2}{6} \times 1 - \frac{2}{6} \times 1$

$= 1 - \frac{1}{3} - \frac{1}{3} - \frac{1}{3}$

$= \boxed{0}$

b) Attribute :- Credit History

Entropy $(S_{Good}) = 0$

Entropy $(S_{Poor}) = 0$

$Gain(S, CH) = 1 - 0 = \boxed{1}$

Total Marks of Question no.		Examiner	
		Moderator	
		Re-Assessor	

Space for Marks	Question No.	START WRITING HERE
	c)	Attribute :- Loan Amount

$$Gain(S, LA) = 1 - \frac{2}{6} \times 0 - \frac{2}{6} \times 1 - \frac{2}{6} \times 0$$

$$= \boxed{0.667}$$

Highest IG is of Credit History. So this will be root node.

